

Thesis/Project Title (<10 words):

GhostGuard: Counterfactual Risk-Calibrated Cooperative Perception

Synopsis of Proposed Research (<=250 words):

Cooperative perception allows connected vehicles and roadside units to share detections that reveal objects outside an ego vehicle's field of view. However, delayed, misaligned, uncertain, or fabricated messages can also introduce duplicate detections and false-positive ghost objects. Current research improves communication efficiency, hard-object coverage, uncertainty weighting, consensus, or sender trust. These signals are useful but indirect: they do not measure whether admitting one object message actually makes the ego vehicle's final perception safer. A confident, provenance-valid false box may be harmful, while an uncertain remote-only pedestrian may be valuable. This project proposes GhostGuard, a late-fusion framework that learns the counterfactual marginal utility of each object message. During training, messages will be added to sampled message subsets and labeled by the resulting change in a safety-weighted perception risk. A lightweight gate will use detector, geometric, temporal, and cross-agent evidence to accept, soft-fuse, quarantine, or reject each candidate. Conformal risk calibration will define an abstention region when evidence is insufficient. Evaluation will separate benign faults from fabricated messages and will measure both harm reduction and preservation of remote-only detections.

Body of the Proposal

A. Introduction

Cooperative perception reveals vehicles and vulnerable road users hidden by occlusion or distance. Yet pose error, delay, association mistakes, sensor degradation, and fabricated reports can create duplicates, displacement, misses, or ghost objects. Fusion or Confusion? demonstrates ghosts after V2X fusion, and RCP-Bench documents failures under environmental, sensor, and temporal corruption [8], [9]. The central question is not how to fuse more data, but whether each received object should affect the ego output.

A.1 Current Research

FPV-RCNN selects 3D keypoint features and applies maximum-consensus localization correction [12]; FocalComm mines hard instances and query-weights exchanged features, improving compressed pedestrian detection [13]. They show that selective communication can preserve useful evidence under bandwidth limits.

Reliability methods estimate how strongly to trust data: BELT uncertainty, CoDynTrust asynchronous-region uncertainty, UECP LiDAR-density uncertainty, and CoDS inter-agent disagreement [1]-[4]. ROBOSAC uses ego consensus; MATE models agent/track trust over time; and Collaborative Anomaly Detection checks cross-vehicle occupancy consistency against fabrication [5]-[7].

Adjacent work adds two relevant ideas. VGMR uses cross-modal agreement and discrepancy for context-conditioned pre-fusion refinement [14]. A provenance-aware AV framework combines cryptography, reputation, a transformer, and accept/quarantine/reject thresholds [15]. These establish that generic value gating and AV-message abstention are not new.

A.2 Limitations of Current Research

- **Selection is optimized for efficiency or salience, not realized harm.** FPV-RCNN is synthetic-only and represents delay through message size; FocalComm is LiDAR-only and leaves

adverse weather, multimodal fusion, and convergence analysis open [12], [13]. Both optimize compact or hard-instance features, not whether one candidate increases ego risk; fabricated-message tests are outside their reported evaluations.

- **Uncertainty, agreement, and sender trust remain proxy signals.** A confident fabricated box may harm, while a low-agreement remote-only pedestrian may help. Coordinated attackers can agree, and defenses requiring ego or benign-agent confirmation may discard the occlusion benefit [1]-[7].
- **Existing contextual value estimation is too coarse for object messages.** VGMR pools modality summaries, which its authors note may miss localized conflicts such as occlusion or frame drops; it is evaluated on non-driving tasks and controlled corruptions [14]. It neither labels individual object messages counterfactually nor calibrates perception harm.
- **Existing quarantine decisions do not measure perception utility.** The provenance framework uses engineered synthetic records without temporal correlations, reports a 21.67% malicious-update miss rate in its test setting, and leaves adaptive attacks and distribution shift open [15]. Its fixed thresholds judge provenance/update records, not the change an object report causes in detection risk.

A.3 Research Gap

The gap is therefore specific: no identified method labels one cooperative object message by its context-dependent marginal change in ego perception risk, then calibrates an abstaining policy against that harm. GhostGuard combines coalition-sampled counterfactual utility, object-level selective fusion, conformal risk control, and separate benefit/harm evaluation across benign faults and fabrication. This is narrower than claiming the first trustworthy, value-aware, or abstaining fusion system.

Research question: Can message-level marginal utility reduce cooperative harm while preserving valuable remote-only detections?

B. Specific Proposed Research and Why Important/Exciting

GhostGuard begins at object-level late fusion, where decisions are interpretable and counterfactual reruns are manageable. It will classify each candidate as helpful, redundant, harmful, or unresolved, then accept, soft-fuse, quarantine, or reject it. The hypothesis is that measured safety-risk change predicts harm better than confidence, agreement, or reputation alone. The aim is evidence for this hypothesis, not a universal safety guarantee. MOT-CUP already applies conformal uncertainty to cooperative tracking [10]; GhostGuard contributes the complete message-utility and calibrated-harm pipeline.

C. Methodology and Why Innovative/Creative

OpenCOOD with PointPillars late fusion will serve as the initial platform. Let $F(S)$ be the fused output from message set S , and let $R(F(S), y)$ combine missed-object, ghost-object, localization, duplicate, and driving-corridor losses.

$$U(m | S) = R(F(S), y) - R(F(S \cup \{m\}), y)$$

Positive, near-zero, and negative utility indicate helpful, redundant, and harmful messages. Because interactions matter, training samples coalitions and estimates $U(m) = E_S[U(m | S)]$, a Shapley-inspired approximation rather than exact Shapley value or causal identification. The detector stays frozen initially; labels require only association and fusion reruns.

A lightweight gate predicts utility, harm probability, and uncertainty from confidence, class/size/range, age, pose uncertainty, association residuals, agreement, temporal persistence, path relevance, and optional sender provenance or history. Metadata is not assumed trustworthy. Actions are

accept, soft-fuse, quarantine for temporal confirmation, or reject; quarantine protects unsupported but real remote-only objects.

Conformal risk control [11] will calibrate nested action thresholds against cooperative harm on held-out data; risk-coverage will also be tested under shift. Baselines are ego-only, naive late fusion, confidence/geometry, BELT-Fusion, ROBOSAC, MATE when reproducible, and CAD. OPV2V/Adv-OPV2V cover benign faults and spoofing/removal; V2V4Real or DAIR-V2X tests transfer. Metrics are 3D AP, remote-only recall, Benefit/Harm Rates, malicious-message acceptance, risk-coverage, bytes, and latency.

D. Milestones and Timeline

- Months 1-2: Reproduce late fusion/baselines; finalize risk and perturbations.
- Months 3-5: Generate coalition labels; train and ablate the utility gate.
- Months 6-8: Calibrate actions; evaluate benign faults and fabrication attacks.
- Months 9-12: Test real data, profile latency, and complete the report, presentation, and manuscript draft.

E. Anticipated Outcome

Deliverables are reproducible code, a cooperative benefit/harm benchmark, quantitative ablations, an adviser-approved report, and a CoE/SJSU presentation. If results support the hypothesis, a manuscript will also be prepared.

G. References

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